

tests/lib/evidence.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** Active sourceable data-plane anti-bluff evidence library for the VPN-aware proxy extension. The committed canonical oracle (§11.4.69 / §11.4.107).

Companion (§11.4.18) to the in-source documentation block at the top of the script. This is a **sourced library**, not a runnable script — source it and call its functions.

Overview / Purpose

A sourceable library of DATA-PLANE evidence helpers. Every helper returns 0 (PASS) only when it can cite CAPTURED data-plane evidence that the user-visible behaviour really works. Control-plane / config / "absence-of-error" signals are NEVER accepted as proof (§11.4 / §11.4.69 / §11.4.107). It is the committed canonical oracle that the dynamic-mode analyzers under tests/dynamic/ source read-only and delegate to (reuse, never reimplement — §11.4.74).

Public function contract

Function	PASS (rc 0) when...
ab_pass_with_evidence <desc> <evidence_path>	the cited evidence artefact EXISTS and is NON-EMPTY (else FAIL, rc 1). The canonical §11.4.69 PASS helper.
ab_skip_with_reason <desc> <reason>	<reason> is in the §11.4.69 closed set (geo_restricted, operator_attended, hardware_not_present, topology_unsupported, network_unreachable_external, feature_disabled_by_config) → honest SKIP (rc 0); any other reason → FAIL (rc 2).
wg_transfer_delta <iface> <before> <after>	the WireGuard tx delta > 0 AND rx delta > 0 across two wg show <if> transfer snapshots; flat/decreasing counters FAIL.

Function	PASS (rc 0) when...
assert_egress_ip <proxy_url> <expected_exit> <host_real>	egress observed THROUGH the proxy == <expected_exit> AND != <host_real>; egress==host is the §15 bluff → FAIL; host IP unknown/empty/ non-IP-shaped → OPERATOR-BLOCKED (rc 2), never fail-open (§11.4.68).
assert_cache_hit <access_log> <url>	the Squid access.log carries a TCP_*HIT result code for <url> (whole-field, not substring).
assert_graceful_503 <proxy_url> <target> <pid_before> <pid_after>	HTTP 503 + non-blank branded body (marker, default tunnel) + PID unchanged (proxy did not crash/ restart).
assert_no_leak <capture_file>	zero target packets escaped the real uplink during the tunnel-down window (tcpdump 0 packets, or /proc/ net/dev tx-packet delta == 0, or zero IP lines).

Supporting public helpers: `procdev_field <file> <iface> <idx>`
(pure /proc/net/dev field accessor), `proxy_conn_verdict <proxy>`
`<direct> <expected> <listening>` (PURE connectivity classifier →
PASS/FAIL/SKIP:...), `port_is_listening <port>` (live listen probe).
Helpers prefixed `_evidence_ / _code_in / _evidence_ip_shaped` are
internal.

Usage

```
. tests/lib/evidence.sh           # source it; do NOT
                                execute
```

```
assert_egress_ip "$PROXY_URL" "$EXPECTED_EXIT" "$HOST_REAL_IP"
ab_pass_with_evidence "cache hit" "$access_log"
ab_skip_with_reason "stress" "topology_unsupported"
```

Unit-test seams (§11.4.27 — stubs only in the unit layer)

Env var	Effect
EVIDENCE_OBSERVED_IP_FILE	assert_egress_ip reads the observed egress IP from this file instead of live curl.
EVIDENCE_503_CODE_OVERRIDE	assert_graceful_503 uses this HTTP code instead of live curl.
EVIDENCE_503_BODY_FILE	

Env var	Effect
	assert_graceful_503 reads the response body from this file.
EVIDENCE_503_BODY_MARKER	branded marker the 503 body must contain (default tunnel, case-insensitive).
EVIDENCE_LEAK_IFACE	real-uplink iface for the /proc/net/dev no-leak delta (default eth0).
EVIDENCE_IP_ECHO_URL	IP-echo endpoint for the live egress probe (default https://icanhazip.com).
EVIDENCE_CURL_TIMEOUT	curl --max-time for live probes (default 15).

Inputs

Captured artefacts (wg transfer snapshots, Squid access.log, 503 body + PID pair, tcpdump capture / /proc/net/dev snapshots) plus, for the live probes, real curl through the proxy. Unit-test seams (above) feed fixtures with no network.

Outputs

One structured verdict line per call on stdout:

```
PASS: <desc> [evidence: <path-or-detail>]
FAIL: <desc> [reason: <why>]
SKIP: <desc> [reason: <closed-set-reason>]
OPERATOR-BLOCKED: <desc> [reason: ...]
```

Return code: 0 = PASS / valid-SKIP, 1 = FAIL, 2 = invalid SKIP / OPERATOR-BLOCKED.

Side-effects

- The live probes (assert_egress_ip, assert_graceful_503) run curl through the proxy in real use; assert_graceful_503 may create + remove a mktmp body file. port_is_listening inspects the local network stack.
- The pure classifiers/parsers (ab_*, wg_transfer_delta, assert_cache_hit, assert_no_leak, procdev_field, proxy_conn_verdict) have no side-effects.

Dependencies

POSIX sh, awk, grep, tr; curl (live probes only); ss/netstat (port_is_listening only). POSIX-clean — parses under sh -n AND bash -n (§11.4.67); no bash-only constructs ([[]], <<<, arrays, >(), \${v^^}).

Edge cases

- **assert_egress_ip fail-open guard (§11.4.68, findings F7/F-1)** — when host_real is empty, unknown, non-IP-shaped garbage, or a loopback/ unspecified sentinel, the egress!=host half cannot be evaluated: a definitively-wrong exit still FAILs, but an otherwise-good result returns OPERATOR-BLOCKED (rc 2), NEVER a fail-open PASS/SKIP.
- **wg_transfer_delta** — a recent handshake with flat counters is control-plane-green / data-plane-dead → FAIL.
- **assert_cache_hit** — URL-specific: a MISS line for the same URL does NOT satisfy it; the HIT token must be the result-code field, not text inside the URL.
- **proxy_conn_verdict** — a positive DIRECT signal out-ranks the port probe, so a crashed proxy on a working host FAILs rather than fail-opening to SKIP (§11.4.68); a site outage SKIPS rather than false-FAILing (§11.4.1).
- **_code_in** matches whole tokens only (20 does not match 200).

Related scripts

- tests/lib/evidence_selftest.sh — TAP self-test proving every parser is correct AND fails on its negative fixture (§1.1).
- tests/dynamic/lib/analyzer_common.sh — sources this library read-only; the six analyzers under tests/dynamic/analyzers/*.sh delegate to these helpers.
- tests/regression/assert_egress_ip_host_unknown_test.sh — the F7 fail-open regression guard.
- Design §13/§14 (docs/superpowers/specs/2026-06-30-vpn-aware-proxy-extension-design.md); research report C; Constitution §11.4.68 / §11.4.69 / §11.4.107.

Last verified

2026-07-01 — documented against the library source; sh -n / bash -n parse-clean. Every parser is exercised (no network, via fixtures + seams) by evidence_selftest.sh; the live curl probes run against the real dynamic stack in P10.